Calcurion Maths Worksheets

GCSE Mathematics

Topic: Circle Theorems

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. The angle in a semicircle is a ..... angle. **[1]**

2. The angle at the centre is ..... the angle at the circumference when subtended by the same arc. **[2]**

3. Angles in the same segment are ..... **[1]**

4. Opposite angles in a cyclic quadrilateral sum to ..... **[2]**

5. The angle between a tangent and a radius is ..... **[1]**

6. Two tangents from an external point to a circle are ..... **[2]**

7. The alternate segment theorem states that the angle between the tangent and a chord is equal to the angle in the ..... segment. **[2]**

8. In a circle, the angle at the centre is 80°. Find the angle at the circumference subtended by the same arc. **[2]**

9. In a circle, the angle at the circumference is 35°. Find the angle at the centre subtended by the same arc. **[2]**

10. A cyclic quadrilateral has angles 80°, 100°, $x°$ and $y°$. Given opposite angles are 80° and $x°$, find x. **[2]**

11. A tangent touches a circle at P and a radius is drawn to P. The angle between the tangent and a chord is 40°. What is the angle in the alternate segment? **[2]**

12. Two tangents from a point touch a circle at A and B. The angle between the tangents is $50°$. Find the angle AOB where O is the centre. **[3]**

13. An angle in a semicircle is $90°$. If one of the other angles in the triangle is $35°$, find the third angle. **[2]**

14. The angle at the centre is $110°$. Find the angle at the circumference in the same segment. **[2]**

15. In a cyclic quadrilateral $ABCD$, $\angle A = 75°$ and $\angle B = 110°$. Find $\angle C$ and $\angle D$. **[3]**

Section B – Hard (Q16–25)

16. O is the centre of a circle. $\angle AOB = 130°$. Find the angle at the circumference subtended by arc AB. **[2]**

17. A tangent at P meets a chord PQ and the angle between them is $55°$. Find the angle in the alternate segment. **[2]**

18. Two tangents from T touch the circle at A and B. $\angle ATB = 65°$. Find $\angle AOB$ where O is the centre. **[4]**

19. In a circle with centre O, $\angle BAC = 40°$ where A is on the circumference and BC is a diameter. Find $\angle BOC$. **[3]**

20. Prove that the angle in a semicircle is a right angle (brief reasoning). **[4]**

21. A cyclic quadrilateral has exterior angle $70°$. What is the opposite interior angle? **[3]**

22. Tangents from P touch at A and B. $\angle APB = 40°$. Find $\angle OAB$ where O is the centre. **[4]**

23. The angle at the centre is twice the angle at the circumference. If the angle at the circumference is x, write an expression for the angle at the centre. **[2]**

24. In a circle, chord AB subtends 70° at the centre and x° at the circumference in the major segment. Find x. **[3]**

25. Explain why opposite angles of a cyclic quadrilateral sum to 180°. **[4]**

End of Worksheet • *Show all working clearly.*